Supplemental Online Information
Expert Elicitation: Using the Classical Model to Validate Experts' Judgments

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APPLICATIONS OF THE CLASSICAL MODEL

Table 1. Thirty-three applications of the Classical Model were conducted from 2006-March 2015.

Study name	Performed by	Contracted by	Subject
Arkansas	Center for Disease Dynamics, Economics & Policy	Robert Wood Johnson Foundation	Grant effectiveness and child health insurance enrollment (Laxminarayan et al. 2012)
Florida			
Illinois			
Nebraska			
Washington			
CoveringKids			
Tobacco			Grant effectiveness and tobacco control (Colson, Cooke, and Laxminarayan 2016)
Obesity			Grant effectiveness and childhood obesity
Fistula		Disease Control Priorities Project, 3rd Edition	Effectiveness of obstetric fistula repair (Colson, Adhikari, et al. 2015)
San_Diego			Effectiveness of surgery
CDC_ROI		Centers for Disease Control and Prevention	Return on investment in public health tracking (Colson, Cohen, et al. 2015)
Create	Vicki Bier	University of Wisconsin and University of Southen California	Terrorism
Erie_Carp	University of Notre Dame	National Oceanic and Atmospheric Administration and Environmental Protection Agency (EPA)	Establishment of Asian Carp in Lake Erie (Cooke et al. 2014; Wittmann et al. 2014, 2015; Zhang et al. 2016)
GL_NIS			Costs of invasive species in Great Lakes (Rothlisberger et al. 2010, 2012)
Nremoval	University of Maryland	EPA and University of Maryland	Nitrogen removal in Chesapeake Bay watershed (Koch, Febria, et al. 2015; Walsh 2015; Koch, Filoso, et al. 2015)
Hemophilia	University Medical Center Utrecht	University Medical Center Utrecht	Optimal treatment of patients with severe haemophilia (Fischer, Lewandowski, and Janssen 2013)

ATCEP	TU Delft	National Institute for Public Health and the Environment	Air traffic controller human error
FCEP			Flight crew human error
Daniela		Brand Preventie	Fire prevention and control (Hanea 2009)
Liander		Liander	Underground cast iron gas-lines (Forys, Kurowicka, and Peppelman 2013)
Arsenic	Willy Aspinall	University of Cambridge	Air quality levels for arsenic
Biol_Agent		Health Protection England	Human dose-response curves for bioterror agents
CWD		University of Ottawa	Infection transmission risks: chronic wasting disease (cwd) from deer to humans (Tyshenko et al. 2011; Tyshenko, ElSaadany, Oraby, Darshan, et al. 2012)
eBPP		PrioNet	Xenotropic murine leukemia virus-related virus (XMRV) blood/tissue infection transmission risks (Tyshenko, ElSaadany, Oraby, Laderoute, et al. 2012)
IceSheets		University of Bristol, British Antarctic Survey, and Ice2sea.eu Project	Contribution to sea level rise from ice sheets melting due to global warming (Bamber and Aspinall 2013)
PHAC		University of Ottawa	Additional CWD factors
Sheep		University of Bristol, Bristol Environmental Risk Research Centre	Risk management policy for sheep scab control
SPEED		Istituto Nazionale di Geofisica e Vulcanologia (INGV), Italy	Volcano hazards at Vesuvius & Campi Flegrei, Italy (Bevilacqua et al. 2015; Neri et al. 2015)
TdC		Natural Environment Research Council, Economic & Social Research Council, and British Geological Survey	Volcano hazards (Tristan da Cunha) (Hicks et al. 2014)
Effusive_Erupt		UK Government	Icelandic fissure eruptions: source characterization (Loughlin et al. 2013)
TOPAZ		Nuclear Waste Management Organization of Japan (NUMO) and Obayashi Corp	Tectonic hazards for radwaste siting in Japan (Scourse, Aspinall, and Chapman 2015)

Gerstenberger	Matthew Gerstenberger	Natural Hazards Research Platform/GNS Science New Zealand	Canterbury Seismic Hazard Model (Gerstenberger et al. 2014, 2013; Christophersen, Nicol, and Gerstenberger 2011)
Goodheart	Benjamin Goodheart	Embry-Riddle	Airport safety

Source: Raw data for all thirty-three applications is available at the homepage of one of the authors: www.rogermcooke.net.

Notes: This dataset excludes two Classical Model studies conducted 2006-March 2015. One is an ongoing elicitation at the Montserrat Volcano Observatory (Aspinall et al. 2002; Wadge and Aspinall 2014). The second is a massive elicitation commissioned by the World Health Organization, involving seventy-two experts combined into 134 panels of overlapping experts (Aspinall et al. 2016; World Health Organization 2016; Hald et al. 2016). Both studies provided large amounts of data that are useful for examining expert performance and combination. However, they are not included here because they contain many closely related panels with overlapping groups of experts, which makes it impossible to treat each individual panel as an independent study.

REFERENCES

Aspinall, Willy P., and Roger M. Cooke. 2013. 'Quantifying Scientific Uncertainty from Expert Judgement Elicitation'. In *Risk and Uncertainty Assessment for Natural Hazards*, edited by Jonathan Rougier, Steve Sparks, and Lisa Hill, 64–99. Cambridge, UK: Cambridge University Press.

Aspinall, Willy P., Roger M. Cooke, Arie H. Havelaar, Sandra Hoffmann, and Tine Hald. 2016. 'Evaluation of a Performance-Based Expert Elicitation: WHO Global Attribution of Foodborne Diseases'. *PLoS ONE* 11 (3): e0149817. doi:10.1371/journal.pone.0149817.

Aspinall, Willy P., S. C. Loughlin, F. V. Michael, A. D. Miller, G. E. Norton, K. C. Rowley, R. S. J. Sparks, and S. R. Young. 2002. 'The Montserrat Volcano Observatory: Its Evolution, Organization, Role and Activities'. *Geological Society, London, Memoirs* 21 (1): 71–91. doi:10.1144/GSL.MEM.2002.021.01.04.

Bamber, J. L., and W. P. Aspinall. 2013. 'An Expert Judgement Assessment of Future Sea Level Rise from the Ice Sheets'. *Nature Climate Change* 3 (4): 424–27. doi:10.1038/nclimate1778.

Bamber, J. L., W. P. Aspinall, and R. M. Cooke. 2016. 'A Commentary on "How to Interpret Expert Judgment Assessments of Twenty-First Century Sea-Level Rise" by Hylke de Vries and Roderik SW van de Wal'. *Climatic Change* 137 (3–4): 321–28. doi:10.1007/s10584-016-1672-7.

Bevilacqua, Andrea, Roberto Isaia, Augusto Neri, Stefano Vitale, Willy P. Aspinall, Marina Bisson, Franco Flandoli, et al. 2015. 'Quantifying Volcanic Hazard at Campi Flegrei Caldera (Italy) with Uncertainty Assessment: 1. Vent Opening Maps'. *Journal of Geophysical Research: Solid Earth* 120 (4): 2309–2329.

Burgman, Mark A., Marissa McBride, Raquel Ashton, Andrew Speirs-Bridge, Louisa Flander, Bonnie Wintle, Fiona Fidler, Libby Rumpff, and Charles Twardy. 2011. 'Expert Status and Performance'. *PLoS ONE* 6 (7): e22998. doi:10.1371/journal.pone.0022998.

Christophersen, A., A. Nicol, and Matthew C. Gerstenberger. 2011. 'The Feasibility of Using Seed Questions for Weighting Expert Opinion in CCS Risk Assessment'. RPT11-2868. CO2CRC. Canberra, Australia: Cooperative Research Centre for Greenhouse Gas Technologies.

Colson, Abigail R., Sweta Adhikari, Ambereen Sleemi, and Ramanan Laxminarayan. 2015. 'Quantifying Uncertainty in Intervention Effectiveness with Structured Expert Judgement: An Application to Obstetric Fistula'. *BMJ Open* 5 (6): e007233. doi:10.1136/bmjopen-2014-007233.

Colson, Abigail R., Mark A. Cohen, Sudha Regmi, Arindam Nandi, Ramanan Laxminarayan, and Molly K. Macauley. 2015. 'Structured Expert Judgment for Informing the Return on Investment in Surveillance: The Case of Environmental Public Health Tracking'.

SSRN Scholarly Paper ID 2704189. Rochester, NY: Social Science Research Network. http://papers.ssrn.com/abstract=2704189.

Colson, Abigail R., and Roger M. Cooke. 2017. 'Cross Validation for the Classical Model of Structured Expert Judgment'. *Reliability Engineering & System Safety* 163 (July): 109–20. doi:10.1016/j.ress.2017.02.003.

Colson, Abigail R., Roger M. Cooke, and Ramanan Laxminarayan. 2016. 'Estimating the Attributable Impact of a Charitable Foundation's Programs Using Structured Expert Judgment'. Working Paper.

Cooke, Roger M., Susie ElSaadany, and Xinzheng Huang. 2008. 'On the Performance of Social Network and Likelihood-Based Expert Weighting Schemes'. *Reliability Engineering & System Safety*, Expert Judgement, 93 (5): 745–56. doi:10.1016/j.ress.2007.03.017.

Cooke, Roger M., Marion E Wittmann, David M Lodge, John D Rothlisberger, Edward S Rutherford, Hongyan Zhang, and Doran M Mason. 2014. 'Out-of-Sample Validation for Structured Expert Judgment of Asian Carp Establishment in Lake Erie'. *Integrated Environmental Assessment and Management* 10 (4): 522–28. doi:10.1002/ieam.1559.

DeGroot, Morris H. 1974. 'Reaching a Consensus'. *Journal of the American Statistical Association* 69 (345): 118–21. doi:10.1080/01621459.1974.10480137.

Fischer, K., D. Lewandowski, and M. P. Janssen. 2013. 'Estimating Unknown Parameters in Haemophilia Using Expert Judgement Elicitation'. *Haemophilia: The Official Journal of the World Federation of Hemophilia* 19 (5): e282-288. doi:10.1111/hae.12166.

Forys, Monika B., Dorota Kurowicka, and Bas Peppelman. 2013. 'A Probabilistic Model for a Gas Explosion Due to Leakages in the Grey Cast Iron Gas Mains'. *Reliability Engineering & System Safety* 119 (November): 270–79. doi:10.1016/j.ress.2013.06.034.

Gerstenberger, Matthew C., A. Christophersen, R. Buxton, G. Allinson, W. Hou, G. Leamon, and A. Nicol. 2013. 'Integrated Risk Assessment for CCS'. *Energy Procedia*, GHGT-11 Proceedings of the 11th International Conference on Greenhouse Gas Control Technologies, 18-22 November 2012, Kyoto, Japan, 37: 2775–82. doi:10.1016/j.egypro.2013.06.162.

Gerstenberger, Matthew C., Graeme McVerry, David Rhoades, and Mark Stirling. 2014. 'Seismic Hazard Modeling for the Recovery of Christchurch'. *Earthquake Spectra* 30 (1): 17–29. doi:10.1193/021913EQS037M.

Hald, Tine, Willy Aspinall, Brecht Devleesschauwer, Roger Cooke, Tim Corrigan, Arie H. Havelaar, Herman J. Gibb, et al. 2016. 'World Health Organization Estimates of the Relative Contributions of Food to the Burden of Disease Due to Selected Foodborne Hazards: A Structured Expert Elicitation'. *PLOS ONE* 11 (1): e0145839. doi:10.1371/journal.pone.0145839.

Hanea, D. M. 2009. 'Human Risk of Fire: Building a Decision Support Tool Using Bayesian Networks'. Dissertation, Delft, The Netherlands: TU Delft. http://repository.tudelft.nl/view/ir/uuid%3A8c7e167e-ee61-488e-afef-9ad7723f4959/.

Hicks, A., J. Barclay, P. Simmons, and S. Loughlin. 2014. ‘An Interdisciplinary Approach to Volcanic Risk Reduction under Conditions of Uncertainty: A Case Study of Tristan Da Cunha’. *Nat. Hazards Earth Syst. Sci.* 14 (7): 1871–87. doi:10.5194/nhess-14-1871-2014.

Hora, Stephen C., Benjamin R. Fransen, Natasha Hawkins, and Irving Susel. 2013. ‘Median Aggregation of Distribution Functions’. *Decision Analysis* 10 (4): 279–91. doi:10.1287/deca.2013.0282.

Koch, Benjamin J., Catherine M. Febria, Roger M. Cooke, Jacob D. Hosen, Matthew E. Baker, Abigail R. Colson, Solange Filoso, et al. 2015. ‘Suburban Watershed Nitrogen Retention: Estimating the Effectiveness of Stormwater Management Structures’. *Elementa: Science of the Anthropocene* 3 (July): 000063. doi:10.12952/journal.elementa.000063.

Koch, Benjamin J., Solange Filoso, Roger M. Cooke, Jacob D. Hosen, Abigail R. Colson, Catherine M. Febria, and Margaret A. Palmer. 2015. ‘Reply to Comment on “Suburban Watershed Nitrogen Retention: Estimating the Effectiveness of Stormwater Management Structures” by Koch et Al. (Elem Sci Anth 3:000063, July 2015)’. *Elementa: Science of the Anthropocene* 3 (December): 000079. doi:10.12952/journal.elementa.000079.

Laxminarayan, Ramanan, Abigail R. Colson, Sweta Adhikari, and Roger M. Cooke. 2012. ‘The Value of Giving, Part 2’. Princeton, NJ: Princeton University.

Lichtendahl, Kenneth C., Yael Grushka-Cockayne, and Robert L. Winkler. 2013. ‘Is It Better to Average Probabilities or Quantiles?’ *Management Science* 59 (7): 1594–1611. doi:10.1287/mnsc.1120.1667.

Loughlin, S. C., Willy P. Aspinall, C. Vye-Brown, P. J. Baxter, C. Braban, M. Hort, A. Schmidt, T. Thordarson, and C. Witham. 2013. ‘Large-Magnitude Fissure Eruptions in Iceland: Source Characterisation’. BGS Open File Report OR/12/098. British Geological Survey. http://www.bgs.ac.uk/research/volcanoes/LakiEruptionScenarioPlanning.html.

Neri, Augusto, Andrea Bevilacqua, Tomaso Esposti Ongaro, Roberto Isaia, Willy P. Aspinall, Marina Bisson, Franco Flandoli, et al. 2015. ‘Quantifying Volcanic Hazard at Campi Flegrei Caldera (Italy) with Uncertainty Assessment: 2. Pyroclastic Density Current Invasion Maps’. *Journal of Geophysical Research: Solid Earth* 120 (4): 2014JB011776. doi:10.1002/2014JB011776.

Rothlisberger, John D., David C. Finnoff, Roger M. Cooke, and David M. Lodge. 2012. ‘Ship-Borne Nonindigenous Species Diminish Great Lakes Ecosystem Services’. *Ecosystems* 15 (3): 1–15. doi:10.1007/s10021-012-9522-6.

Rothlisberger, John D, David M Lodge, Roger M. Cooke, and David C Finnoff. 2010. ‘Future Declines of the Binational Laurentian Great Lakes Fisheries: The Importance of Environmental and Cultural Change’. *Frontiers in Ecology and the Environment* 8 (5): 239–44. doi:10.1890/090002.

Scourse, E., W. P. Aspinall, and N. Chapman. 2015. ‘Using Expert Elicitation to Characterise Long-Term Tectonic Risks to Radioactive Waste Repositories in Japan’. *Journal of Risk Research* 18 (3): 364–77. doi:10.1080/13669877.2014.971334.

Tyshenko, Michael G, Susie ElSaadany, Tamer Oraby, Shalu Darshan, Willy P. Aspinall, Roger M. Cooke, Angela Catford, and Daniel Krewski. 2011. ‘Expert Elicitation for the Judgment of Prion Disease Risk Uncertainties’. *Journal of Toxicology and Environmental Health. Part A* 74 (2–4): 261–85. doi:10.1080/15287394.2011.529783.

Tyshenko, Michael G., Susie ElSaadany, Tamer Oraby, Shalu Darshan, Angela Catford, Willy Aspinall, Roger Cooke, and Daniel Krewski. 2012. ‘Expert Judgement and Re-Elicitation for Prion Disease Risk Uncertainties’. *International Journal of Risk Assessment and Management* 16 (1–3): 48–77. doi:10.1504/IJRAM.2012.047552.

Tyshenko, Michael G., Susie ElSaadany, Tamer Oraby, Marian Laderoute, Jun Wu, Willy Aspinall, Daniel Krewski, and Peter R. 2012. ‘Xenotropic Murine Leukemia Virus-Related Virus as a Case Study: Using a Precautionary Risk Management Approach for Emerging Blood-Borne Pathogens in Canada’. In *The Continuum of Health Risk Assessments*, edited by Michael G. Tyshenko. InTech. http://www.intechopen.com/books/the-continuum-of-health-risk-assessments/xenotropic-murine-leukemia-virus-related-virus-as-a-case-study-using-a-precautionary-risk-managemen.

Wadge, G., and Willy P. Aspinall. 2014. ‘A Review of Volcanic Hazard and Risk-Assessment Praxis at the Soufrière Hills Volcano, Montserrat from 1997 to 2011’. *Geological Society, London, Memoirs* 39 (1): 439–56. doi:10.1144/M39.24.

Walsh, Christopher J. 2015. ‘Comment on “Suburban Watershed Nitrogen Retention: Estimating the Effectiveness of Stormwater Management Structures” by Koch et Al. (Elem Sci Anth 3:000063, July 2015)’. *Elementa: Science of the Anthropocene* 3 (December): 000077. doi:10.12952/journal.elementa.000077.

Wittmann, Marion E., Roger M. Cooke, John D. Rothlisberger, and David M. Lodge. 2014. ‘Using Structured Expert Judgment to Assess Invasive Species Prevention: Asian Carp and the Mississippi—Great Lakes Hydrologic Connection’. *Environmental Science & Technology* 48 (4): 2150–56. doi:10.1021/es4043098.

Wittmann, Marion E., Roger M. Cooke, John D. Rothlisberger, Edward S. Rutherford, Hongyan Zhang, Doran M. Mason, and David M. Lodge. 2015. ‘Use of Structured Expert Judgment to Forecast Invasions by Bighead and Silver Carp in Lake Erie’. *Conservation Biology* 29 (1): 187–97. doi:10.1111/cobi.12369.

World Health Organization, Foodborne Epidemiology Reference Group, Source Attribution Task Force. 2016. ‘Research Synthesis Methods in an Age of Globalized Risks: Lessons from the Global Burden of Foodborne Disease Expert Elicitation’. *Risk Analysis* 36 (2): 191–202. doi:10.1111/risa.12385.

Zhang, Hongyan, Edward S. Rutherford, Doran M. Mason, Jason T. Breck, Marion E. Wittmann, Roger M. Cooke, David M. Lodge, John D. Rothlisberger, Xinhua Zhu, and Timothy B. Johnson. 2016. ‘Forecasting the Impacts of Silver and Bighead Carp

on the Lake Erie Food Web'. *Transactions of the American Fisheries Society* 145 (1): 136–62. doi:10.1080/00028487.2015.1069211.